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## 1 Classic binomial and multinomial identities

**Definition 1.1.** Factorial of  $n$ , denoted by  $n!$ , is the number of ways to put  $n$  distinct objects in a queue.

**Definition 1.2.** Binomial coefficient  $\binom{n}{k}$  is the number of ways to select  $k$  distinct objects from  $n$  distinct objects.

**Definition 1.3.** For  $n = a + b + c$  the multinomial coefficient  $\binom{n}{a,b,c}$  is the number of ways to split  $n$  distinct objects into different boxes, where the first box contains  $a$  objects, the second box contains  $b$  objects, the third box contains  $c$  objects.

One may create multinomial coefficient for any number of boxes :)

**Exercise 1.1.** Is  $\binom{n}{k}$  equal to  $\binom{n}{n-k}$ ?

Is  $\binom{n}{k}$  equal to  $\binom{n}{k,n-k}$ ?

**Exercise 1.2.** Write the product

$$\binom{100}{20} \cdot \binom{80}{50} \cdot \binom{30}{10}$$

as one multinomial coefficient.

**Exercise 1.3.** Write

$$\binom{n}{a,b,c,d}$$

using binomial coefficients.

Write

$$\binom{n}{a,b,c,d}$$

using factorials of  $n$ ,  $a$ ,  $b$ ,  $c$  and  $d$ .

**Idea 1.** Solution strategies for combinatorial identities:

1. Double counting or story proof.
  2. Visual identity (rare gem!).
  3. Evaluation of a polynomial at a special point. Differentiation, integration.
  4. Combine old identities into new.
  5. Guess the answer and verify by induction.
  6. Dark art!
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## Pascal's triangle and rule

**Exercise 1.4.** Recall the Pascal arithmetic triangle. Draw two versions: triangular and rectangular.

**Exercise 1.5.** Pascal's rule. Simplify

$$\binom{n}{k} + \binom{n}{k+1}.$$
$$\binom{n}{a-1, b, c} + \binom{n}{a, b-1, c} + \binom{n}{a, b, c-1}.$$
$$\binom{n}{a-1, b, c, d} + \binom{n}{a, b-1, c, d} + \binom{n}{a, b, c-1, d} + \binom{n}{a, b, c, d-1}.$$

**Exercise 1.6.** Simplify

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$

Recall the Pascal's rule and simplify

$$\binom{2n}{0} + \binom{2n}{2} + \binom{2n}{4} + \cdots + \binom{2n}{2n}.$$

Simplify

$$\binom{2n-1}{0} + \binom{2n-1}{2} + \binom{2n-1}{4} + \cdots + \binom{2n-1}{2n-2}.$$

**Exercise 1.7.** Simplify

$$\sum_{a+b+c=2026} \binom{2026}{a, b, c}.$$
$$\sum_{a+b+c+d=2027} \binom{2027}{a, b, c, d}.$$
$$\sum_{a+b+c=100} \binom{100}{a} \cdot \binom{100-a}{b}.$$
$$\sum_{a+b \leq n} \binom{n}{a} \cdot \binom{n-a}{b}.$$

Here  $a$ ,  $b$  and  $c$  are assumed non-negative integers.

## Hockey-stick identity

**Exercise 1.8.** Hockey-stick identity, Christmas stocking identity, boomerang identity.

Simplify

$$\sum_{n=a}^b \binom{n}{a}.$$
$$1 + 2 + 3 + \cdots + n.$$
$$1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n \cdot (n+1).$$
$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \cdots + n \cdot (n+1) \cdot (n+2).$$

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**Exercise 1.9.** Obtain closed form expressions for:

$$1^2 + 2^2 + \cdots + n^2,$$

$$1^3 + 2^3 + \cdots + n^3,$$

$$1^4 + 2^4 + \cdots + n^4.$$

**Exercise 1.10.** Hockey-sticks plus idea from linear algebra!

Decompose  $n^2 + 2n + 5$  using  $\binom{n}{0}$ ,  $\binom{n}{1}$ ,  $\binom{n}{2}$  as the basis.

Find the sum

$$\sum_{k=2}^{2026} k^2 + 2k + 5.$$

Using similar trick find the sums

$$\sum_{k=1}^n 5k^2 + 4k, \quad \sum_{k=1}^n k^3 + 2k^2.$$

**Exercise 1.11.** Let's start from some number  $\binom{n}{k}$  inside the Pascal's triangle.

In one step we can go either in north-east or north-west direction.

Find the sum of all numbers we can reach.

### Wandermonde's identity

**Exercise 1.12.** Simplify

$$\binom{a}{0} \binom{b}{m} + \binom{a}{1} \binom{b}{m-1} + \binom{a}{2} \binom{b}{m-2} + \cdots + \binom{a}{m} \binom{b}{0}.$$

$$\binom{a}{0} \binom{m}{0} + \binom{a}{1} \binom{m}{1} + \binom{a}{2} \binom{m}{2} + \cdots + \binom{a}{m} \binom{m}{m}.$$

$$\binom{a}{0}^2 + \binom{a}{1}^2 + \binom{a}{2}^2 + \cdots + \binom{a}{a}^2.$$

**Exercise 1.13.** There are  $n = 500$  fishes in the pond,  $t = 30$  of them are tagged. You capture  $c = 5$  fishes at random without replacement. Let the random variable  $Y$  be the number of captured tagged fishes. Find the probabilities  $\mathbb{P}(Y = 0)$ ,  $\mathbb{P}(Y = 1)$ ,  $\mathbb{P}(Y = 2)$ ,  $\mathbb{P}(Y = y)$ .

**Exercise 1.14.** There are  $n = 24$  candies in the jar, 14 with orange flavour and 10 with ananas flavour. You take 6 candies at random. Let  $Y$  be the number of candies with orange flavour you picked. Find the probabilities  $\mathbb{P}(Y = 0)$ ,  $\mathbb{P}(Y = 1)$ ,  $\mathbb{P}(Y = 2)$ ,  $\mathbb{P}(Y = y)$ .

**Exercise 1.15.** A bridge hand (random 13 cards out of 52-card standard deck) is dealt.

What is the probability that the hand contains no spades?

What is the probability that the hand contains exactly 5 hearts?

What is the probability that the hand contains at most two aces?

**Exercise 1.16.** Simplify

$$\sum_{a+b+c=k} \binom{n_1}{a} \binom{n_2}{b} \binom{n_3}{c}.$$

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## Binomial or Taylor or MacLaurin expansion

**Exercise 1.17.** Write the Taylor expansion (binomial expansion) at  $x = 0$  for

$$(1+x)^n, \quad (x+y)^n.$$

**Exercise 1.18.** By plugging some special points to the  $(x+y)^n$  or somewhere else find the sums

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$
$$\sum_{k=0}^n k \binom{n}{k}, \quad \sum_{k=0}^n (k-1) \binom{n}{k}, \quad \sum_{k=0}^n (-1)^k \binom{n}{k}, \quad \sum_{k=0}^n (-1)^{k-1} k \binom{n}{k}$$

Hint: differentiation may help you.

### How that can be?

**Exercise 1.19.** Be brave, really brave!

Take a deep breath and calculate

$$\binom{-1}{3}, \quad \binom{-5}{5}, \quad \binom{1.5}{3}, \quad \binom{-1/2}{4}, \quad \binom{1/2}{k}, \quad \binom{-1/2}{n}.$$

**Exercise 1.20.** Simplify

$$\binom{-1/2}{3} + \binom{-1/2}{4}.$$

Draw the Pascal's triangle starting from  $\binom{-1/2}{0}$ .

Draw the Pascal's triangle starting from  $\binom{1/3}{0}$ .

**Exercise 1.21.** Write the Taylor expansion at  $x = 0$  for

$$\frac{1}{1-x}, \quad (1+x)^{-5}, \quad (1+x)^{-1/2}, \quad (1+3x)^{1/3}, \quad (1-5x)^{1/2}.$$

**Exercise 1.22.** Simplify

$$\binom{10}{0} \binom{-0.5}{7} + \binom{10}{1} \binom{-0.5}{6} + \binom{10}{2} \binom{-0.5}{5} + \cdots + \binom{10}{7} \binom{-0.5}{0}.$$

Hint: for a honest proof consider the binomial expansion of  $(1+x)^1, (1+x)^{-0.5}, (1+x)^{10-0.5}$ .

**Exercise 1.23.** Homage to Newton.

Write the Taylor expansion at  $x = 0$  for  $(1-x^2)^{1/2}$ .

Visually (or analytically if you wish) find the integral

$$\int_{-1}^1 (1-x^2)^{1/2}.$$

Calculate the sums

$$\binom{1/2}{0} - \frac{1}{3} \binom{1/2}{1} + \frac{1}{5} \binom{1/2}{2} + \cdots$$
$$\sum_{k=2}^{\infty} \frac{1}{2k+1} \frac{(2k-3)!!}{(2k)!!}$$

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## Gauss summation trick

**Exercise 1.24.** Simplify

$$\begin{aligned} & 1 + 2 + 3 + \cdots + n; \\ & 1^2 + 2^2 + 3^2 + \cdots + n^2; \\ & \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n}; \\ & \binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \cdots + (n+1)\binom{n}{n}; \\ & \binom{n}{2} + 2\binom{n}{3} + 3\binom{n}{4} + \cdots + (n-1)\binom{n}{n}; \\ & \binom{n}{0} + 3\binom{n}{1} + 5\binom{n}{2} + \cdots + (2n+1)\binom{n}{n}; \\ & \binom{n}{0} - 2\binom{n}{1} + 3\binom{n}{2} - 4\binom{n}{3} + \cdots + (-1)^n(n+1)\binom{n}{n}; \\ & 3\binom{n}{1} + 7\binom{n}{2} + 11\binom{n}{3} + \cdots + (4n-1)\binom{n}{n}; \\ & \binom{n}{1} - 2\binom{n}{2} + 3\binom{n}{3} - 4\binom{n}{4} + \cdots + (-1)^n n\binom{n}{n}; \end{aligned}$$

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